

Real Estate Property Management System

Backend Development Project

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Problem Statement



Key Challenges

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Property searching is often handled through multiple websites, agents, or manual processes.

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Buyers may not have complete and updated information about available properties.

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Comparing properties based on price, location, amenities, and other details can be difficult.

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Property documents, site-visit details, and communication may be managed separately.

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Buyers need better transparency and communication throughout the property-buying process.

Objectives



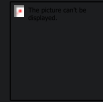
Streamlined Search

Simplify property search and discovery for end users with intelligent tools.



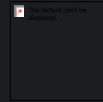
Filter & Compare

Implement efficient filtering and comparison tools for smarter decisions.



Centralized Database

Create a single, reliable database hub for all property listings.



Seamless UX

Deliver a smooth user experience across all platforms and screen sizes.

Proposed System

- 1 A centralized mobile system for managing the property buying process.
- 2 Buyers can easily search and explore available properties.
- 3 Property details such as price, location, area, and amenities are displayed.
- 4 Buyers can filter and compare properties based on their preferences.
- 5 Users can save and manage their favorite properties.



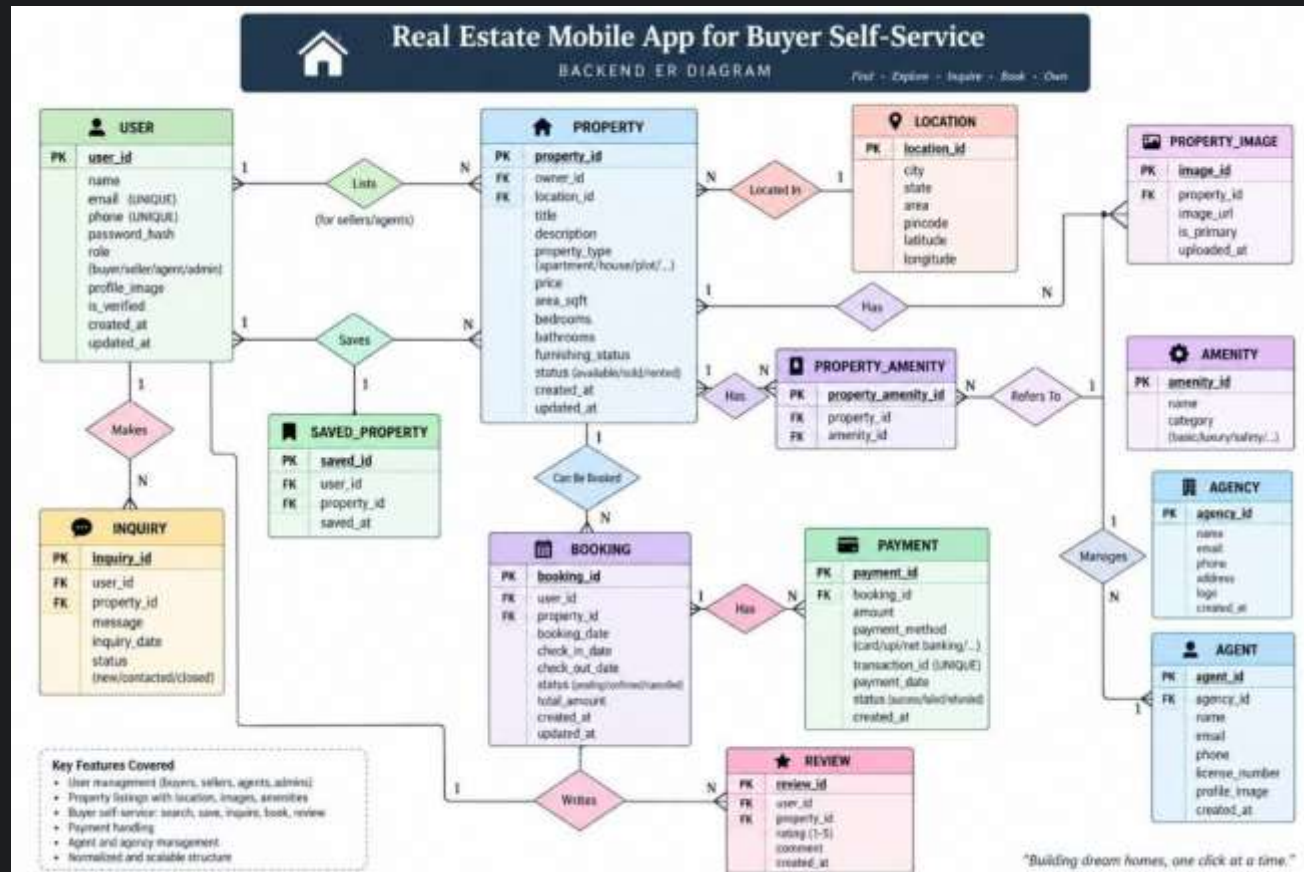
System Modules

Buyer Modules	Property Management Modules
• Dashboard	• Property Listing Management
• Property Search	• Property Information
• Property Listings	• Property Images
• Property Details	• Property Amenities
• Property Filtering	• Property Location
• Saved Properties	• Property Availability
• Property Visit Scheduling	• Visit Management
• My Visits	• Buyer Enquiries
• Buyer Profile	• Property Verification
• Notifications	• Property Status
• Favorites Management	• Featured Properties
• Contact Agent	• Property Updates

Frontend Implementation – Screenshots



ER Diagrams



The ER diagram of EstateLy represents the main entities involved in the buyer self-service real estate system. The Buyer entity interacts with Property through Saved Property and Property Visit entities. Saved Property records the properties bookmarked by buyers, while Property Visit records scheduled visits. The Agent entity represents property consultants who manage properties. Primary and foreign keys establish the relationships between these entities.

Database Design & Schema



User Table

Stores buyer details such as User ID, Name, Email, Phone Number, and Password for managing user profiles.



Property Table

Stores property information including Property ID, Title, Location, Price, Property Type, Bedrooms, Bathrooms, Area, and Image.



Favorites Table

Maintains the properties saved by buyers, containing Favorite ID, User ID, and Property ID.



Visit Table

Stores property visit details such as Visit ID, User ID, Property ID, Visit Date, Visit Time, and Visit Status.



Relationships

A user can save multiple properties and schedule multiple visits. Each property can be saved or visited by multiple users, creating relationships between **User, Property, Favorites, and Visits**.

Database Tables / Schema

Table Name	Primary Key	Important Fields	Description
Users	user_id	name, email, phone, password, role	Stores buyer/user account information and login details.
Properties	property_id	title, location, price, type, bedrooms, bathrooms, area, description, image	Stores details of properties available for buyers.
Amenities	amenity_id	amenity_name, description	Stores available property amenities such as parking, security, Wi-Fi and fitness center.
Property_Amenities	property_amenity_id	property_id, amenity_id	Connects properties with their available amenities.
Favorites	favorite_id	user_id, property_id, saved_date	Stores properties saved by buyers for future reference.
Visits	visit_id	user_id, property_id, visit_date, visit_time, status	Stores property visit/schedule information requested by buyers.
Agents	agent_id	name, phone, email, verification_status	Stores property consultant/agent information.
Property_Agents	property_agent_id	property_id, agent_id	Associates properties with their respective agents.



Project Status & Next Steps

Current Progress

01

Requirements Analysis

Problem scope, buyer requirements, user stories, and frontend system requirements have been identified and documented.

02

Frontend Development

Home page, property listing, property details, login, registration, and buyer dashboard have been implemented using React.

03

UI/UX Design

Modern, responsive real estate interface designed with a premium **EstateLy** theme for easy property discovery and buyer self-service.

04

Frontend Testing

Navigation, property details, saving properties, dashboard updates, and responsive layouts are being tested and refined.

Future Work

- AI-powered property recommendations
- Mobile app development for Android and iOS
- Map integration for property locations
- Payment gateway integration for future booking features
- Property analytics and buyer activity dashboard



Challenges / Issues Faced

- **Responsive Design:** Ensuring the application works properly on desktops, tablets, and mobile screens.
- **Component Integration:** Managing multiple React components and connecting pages smoothly using React Router.
- **State Management:** Handling features such as saved/favorite properties and maintaining their state using browser storage.
- **UI Consistency:** Maintaining a consistent, user-friendly design, layout, icons, colors, and navigation across all pages.

Future Work

- **Backend Integration:** Add a secure backend and database to store users, properties, favorites, and visit details.
- **Advanced Search:** Implement filters for price, location, property type, bedrooms, and amenities.
- **Online Property Visits:** Add appointment booking, calendar integration, and notifications for scheduled visits.
- **Mobile Application:** Convert the web application into a dedicated Android/iOS app for convenient property browsing and buyer self-service.



Conclusion

- **EstateLy** provides a simple and user-friendly platform for buyers to explore properties and view detailed information.
- The application allows users to **search, view, and save properties** according to their preferences.
- Features such as **property details, favorites, buyer dashboard, login, registration, and visit scheduling** improve the overall user experience.
- The responsive frontend design makes the application accessible across **desktop, tablet, and mobile devices**.
- Overall, the project demonstrates how a modern **frontend-based real estate application** can simplify the property-buying and management experience for buyers.



Thank You



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